

sufficient cognitive effects of the expected kind, the audience (Peter) is entitled to attribute to Mary the intention to communicate the thoughts in (11).

However, notice that the utterance (1b) has raised the accessibility of other contextual assumptions as well:

- (12) a. Coke is an energizing drink.
- b. Redbull is an energizing drink.
- c. Tea is an energizing drink.
- d. Peppermint tea is an energizing drink.

Though not as highly accessible as, and hence less manifest than, the assumption *Coffee is an energizing drink*, at least some of them are still very easily accessible, and by inferential processes closely parallel to the one in (11), these in turn lead to the conclusions in (13):

- (13) a. Mary does not want to drink coke.
- b. Mary does not want to drink redbull.
- c. Mary does not want to drink tea.
- d. Mary does not want to drink peppermint tea.

While the interpretation in (11) alone would suffice to make Mary's utterance relevant enough to Peter to be worth the attention, expanding the context slightly by accessing, for instance, (12a) and (12b) will yield more cognitive effects. Assuming that doing so does not obviously violate the audience's estimation of the communicator's abilities and preferences, the audience is licensed to include these assumptions and the inferences they support in their interpretive hypothesis, although with less confidence than the more easily accessible (and hence more relevant and more manifest) assumption *B does not want to drink coffee*. What about the still less accessible (and hence less manifest) assumptions (13c) and (13d)? This depends on the audience's expectations of relevance, in particular on how much processing effort the audience is prepared to tolerate.

So in other words, this analysis suggest that all the possible contextual assumptions linked to the encyclopaedic entries of the concept *ENERGIZING DRINK*, and the contextual implications they give rise to, are becoming more manifest to Peter as a result of processing Mary's utterance following the relevance-theoretic comprehension procedure. They are becoming more manifest to different degrees, as they differ in accessibility. Recall that the manifestness of a piece of information is a measure of the probability that the piece of information will play a causal role in the individual's mental life. This probability itself does not entail that the individual actually mentally represents this piece of information; it merely entails that the individual is capable, in principle, to mentally represent this piece of information and assess its truth. Thus, the high probability that (3a) and (3b) will be accepted by Peter as part of Mary's intended meaning does not itself guarantee that Peter will do so; but it will give Peter a strong incentive for doing so. Sperber and Wilson (1995, p. 199) say that implicatures such as (3a) and (3b) are *strongly communicated* and constitute *strong implicatures*. Peter's incentive for representing implicatures such as (12d) and (13d) is much weaker. Sperber and Wilson (1995, p. 199) speak of *weakly communicated* assumptions and call them

weak implicatures. Obviously, the distinction between strong and weak implicatures is not a categorical one; rather, there is a cline between them.

For the purposes of this paper, we will assume that Peter represents only the implicatures (3a) and (3b) and attributes them to Mary's informative intention, the implicatures for which Peter has the strongest grounds for attributing them to Mary. Still, the weaker implicatures have a certain likelihood to influence Peter's behaviour: realising that Mary declined his offer of a cup of coffee, he may want to offer her something else instead. At this point, the weaker communicated implicatures may influence his alternative offer.

The comprehension process of the implicatures in example (1b) are summarised in table 1.

4 Observing the contours of the comprehension heuristic

4.1 Inference under uncertainty: heuristics and probabilistic models

4.1.1 Communicaton and inference under uncertainty

The philosopher Grice was intrigued by the question why logical connectives in natural languages, like *and* in English, seem to deviate in their meaning from the logical operator $\wedge$, as illustrated by contrast between (14a) and (14b):

- (14) a. John ran to the edge of the cliff and jumped.
- b. John jumped and ran to the edge of the cliff.

Does this show that the meaning of natrual language connectors differs from their counterpart in formal logic so that linguists must discover a special natural language logic (which is presumably more subtle and complicated than formal logic)? Grice (1967) (later published in Grice 1989) argued that this is not the case. Rather, he attributed the difference between (14a) and (14b) to the fact that natural language productions are used by *speakers* as *utterances* in *conversations* with *hearers*, and such social exchanges are governed by a set of *maxims* that are grounded in rationality. These implicitly shared rationality maxims make it possible for speakers to convey meanings that are more subtle than what the semantics of natural language expressions allows for, and to convey their intended meaning with utterances that are more economical than fully literal expressions would be. Implicatures, and in fact various types of implicatures, are a theoretical tool to describe the difference between semantics and the speaker's meaning.

This general account makes the groundbreaking, even monumental, claims:

- (15) Inference is central in the communication of spekaer's meaning:
 - a. Speakers choose utterances on the basis of which conversational maxims rational listeners would expect them to follow, while intending to achieve their goals;
 - b. Listeners comprehend utterances on the basis of what conversational maxims rational communicators would be expected to implement;

Representation	Comment
(a) Mary has said to Peter “I don’t want to drink energizing drinks at this time of day.”	Embedding of the decoded incomplete logical form of Mary’s utterance into a description of Mary’s overtly intentional act.
(b) Mary’s utterance is optimally relevant to Peter.	Expectation raised by the overt nature of Mary’s intentional act and acceptance of the presumption of relevance it conveys.
(c) Mary’s utterance is relevant to Peter by specifying whether or not Mary wants to drink some coffee now at 6 pm.	Fine-tuned expectation of relevance raised by (b) in the light of Peter’s preceding question.
(d) Coffee is an energizing drink.	Encyclopaedic entry of the concept <code>ENERGIZING DRINK</code> . Accepted as an implicit premise, an implicature of Mary’s utterance.
(e) Mary does not want to drink energising drinks at 6 pm.	Proposition derived by reference resolution of the pronoun <i>I</i> and fixing the time reference of the phrase <i>at this time of day</i> to 6 pm, the time of the talk exchange. Accepted as the explicature of Mary’s utterance.
(f) Mary does not want to drink coffee at 6 pm.	Implicit conclusion from (d) and (e). Satisfies expectation (c) and (b). Accepted as an implicature of Mary’s utterance.
(g) Coke is an energising drink.	Other encyclopaedic entry of the concept <code>ENERGIZING DRINK</code> . Serves as another implicit premise.
(h) Mary does not want to drink coke at 6 pm.	Implicit conclusion from (g) and (e). Increases satisfaction of expectation (c) and (b). Accepted as a less manifest implicature of Mary’s utterance.
[The encyclopaedic entries of the concept <code>ENERGISING DRINK</code> may provide other premises similar to (d) and (g)]	[Potentially supporting further implicit conclusions, each less manifest the less accessible they are, and the less processing effort Peter is prepared to invest.]

Table 1: Summary table of implicature comprehension in example (1b)